

SS 465 data from carpenter technology 2

FOR CONDITION **H950 AND H1000**

REFERENCE LINKS

<https://www.carpentertechnology.com/hubfs/7407324/Material%20Safety%20Data%20Sheets/Custom%20465.pdf>

FOR CONDITION H950 AND H1000

Effects of Exposure at Elevated Temperatures on Room Temperature Mechanical Properties—Custom 465 Stainless Bar*

Thermal Stability, 4½" x 1½" Forged Bar

Condition	0.2% Yield Strength		Ultimate Tensile Strength		% Elongation in 4D	% Reduction of Area	Notch Tensile Strength**		Charpy V-Notch Impact		Hardness (HRC)
	ksi	MPa	ksi	MPa			ksi	MPa	ft-lbs	J	
H950	239	1648	255	1758	14	62	367	2530	20	27	49
H950 + 600°F (316°C), 1000 hrs, AC	242	1669	259	1786	16	59	354	2441	12	16	49.5
H950 + 700°F (371°C), 1000 hrs, AC	250	1724	268	1848	14	56	285	1965	11	15	51
H950 + 800°F (427°C), 1000 hrs, AC	253	1744	272	1875	13	54	271	1868	9	12	51.5
H950 + 900°F (482°C), 1000 hrs, AC	211	1455	223	1538	19	67	332	2289	34	46	46
H1000	218	1503	231	1593	16	66	355	2448	38	52	47
H1000 + 600°F (316°C), 1000 hrs, AC	219	1510	232	1600	18	65	354	2441	30	41	47
H1000 + 700°F (371°C), 1000 hrs, AC	226	1558	240	1655	16	65	350	2413	25	34	47.5
H1000 + 800°F (427°C), 1000 hrs, AC	229	1579	245	1689	15	62	347	2392	18	24	48
H1000 + 900°F (482°C), 1000 hrs, AC	210	1448	222	1531	20	66	327	2255	32	43	46

* All longitudinal properties

**Notch tensile strength with $K_t = 10$